



GRADSKOOL PRESENTS • FYQS

CAT Percentages FYQs Booklet

50 Future Year Questions (FYQs)

FYQS

50

FORMAT

MCQ

TOPIC

Percentages

GRADSKOOL — FYQ SERIES FOR CAT PREPARATION

01 Questions (FYQs)

Percentages — 50 Future Year Questions (FYQs)

INSTRUCTIONS

This booklet contains 50 Future Year Questions (FYQs) — original, CAT-style questions engineered to mirror the difficulty and pattern of upcoming exams. Attempt all 50 FYQs in this section before checking the Answer Key. Each question carries equal weight. Recommended time: 2–2.5 minutes per FYQ. Solutions follow the Answer Key section.

01

FYQ · MCQ

A vessel contains V litres of a solution that is 50% acid. First, 20% of the solution is drained and replaced with water. Then, 30% of the resulting mixture is drained and replaced with pure acid. Finally, 25% of this new mixture is drained and replaced with water. If the final amount of acid in the vessel is exactly 130.5 litres, find V .

A) 250

B) 280

C) 300

D) 320

02

FYQ · MCQ

A class has 420 students, consisting only of boys and girls. In the first scenario, 20% of the boys and 25% of the girls leave the class. Subsequently, 12 new boys and 15 new girls join. After these changes, the number of girls is exactly 25% more than the number of boys. In a second scenario, 30% of the original boys and 15% of the original girls leave instead, followed by the same 12 boys and 15 girls joining. In this case, the number of girls exceeds the number of boys by 81. Find the original number of boys.

A) 200

B) 180

C) 150

D) 240

03

FYQ · MCQ

A number is increased by $p\%$, then decreased by $q\%$, where p and q are positive integers with $p+q=50$. If the net result is exactly a 4% increase over the original number, find p .

A) 30

B) 25

C) 35

D) 20

04

FYQ · MCQ

A person's gross monthly salary is S . First, 20% of the salary is deducted as tax. From the amount remaining, a fixed bonus of ₹1,500 is added. Then 15% of this new amount is deducted as insurance. Finally, ₹1,200 is deducted as EMI. The final take-home salary is ₹8,235. However, the person later receives a revised salary structure in which the tax deduction becomes 25% of the gross salary, the bonus becomes ₹2,500, the insurance deduction becomes 10% of the amount after adding the bonus, and the EMI remains unchanged. Under the revised structure, the take-home salary would be ₹9,150. Find S .

- A) ₹10,000
- B) ₹11,500
- C) ₹12,500
- D) ₹12,000

05

FYQ · MCQ

Three friends' incomes are in the ratio 5:6:9. They contribute 10%, 20%, and $r\%$ of their respective incomes toward a shared bill, and this combined contribution equals 22% of their total income. Find r . If their incomes had instead been in the REVERSED ratio 9:6:5 (with the same three contribution rates 10%, 20%, and $r\%$ now applied in that new order), what would the combined contribution be as a percentage of total income?

- A) $r = 25$, new% = 18%
- B) $r = 35$, new% = 15%
- C) $r = 30$, new% = 18%
- D) $r = 30$, new% = 20%

06

FYQ · MCQ

Two factories, A and B, initially produce the same total output. Initially, the output per worker in Factory A is 75% of the output per worker in Factory B, and Factory A has 240 more workers than Factory B. During the following year, the output per worker in Factory A increases by 25%, after which 20% of its workers leave, and the productivity of each remaining worker increases by a further 8%. In Factory B, the output per worker decreases by 10%, the workforce increases by 25%, and then 20% of the resulting workforce leaves. After all these changes, Factory A's total output is exactly 20% greater than Factory B's total output. Find the original number of workers in Factory A.

- A) 960
- B) 840
- C) 720
- D) 1,200

07

FYQ · MCQ

In a survey, 70% of respondents use Product A, 80% use Product B, and 60% use Product C. Find the difference between the MAXIMUM possible percentage using all three products and the MINIMUM possible percentage using all three products.

- A) 50
- B) 45
- C) 60
- D) 40

08

FYQ · MCQ

Two positive integers x and y satisfy the following conditions: when x is decreased by $y\%$, the result is 48, and when y is increased by $x\%$, the result is 72. If x is increased by 20% while y is decreased by 20%, the resulting difference between the two numbers is 64. Find x .

A) 90

B) 72

C) 80

D) 60

09

FYQ · MCQ

A town has a population of 3,000 at the beginning of Year 1. At the end of each year, its population increases by 20%, after which 100 people migrate out. However, at the end of every third year, an additional 200 people migrate out after the regular migration. Find the first year at the end of which the population exceeds 12,000.

A) 8th year

B) 9th year

C) 7th year

D) 10th year

10

FYQ · MCQ

Country A exports a total of T units to Countries B, C, D and E. A fixed 120 units are exported to D, while the remaining exports are divided among B, C and E in the ratio 2:3:1. Country B imports 1,500 units in total, of which 16% are supplied by Country A. Find the total exports T of Country A.

A) 720

B) 900

C) 780

D) 840

11

FYQ · MCQ

A quantity undergoes three successive percentage increases, $a\%$, $b\%$, and $c\%$, where $a+b+c=90$ (all positive). What is the maximum possible overall percentage increase in the quantity?

A) 119.7%

B) 125%

C) 110%

D) 100%

12

FYQ · MCQ

Let N be a positive integer such that $1.25N$ lies strictly between 150 and 200, $0.60N$ lies strictly between 80 and 130, N is a multiple of 6, and N is NOT a multiple of 5. How many values of N satisfy all four

conditions?

- A) 2 values
- B) 4 values
- C) 5 values
- D) 3 values

13

FYQ · MCQ

Two positive numbers A and B satisfy $A \times B = 600$ (their PRODUCT, not their sum), with $A > B$. If A increases by 10% and B increases by 40%, the new difference ($A_{\text{new}} - B_{\text{new}}$) becomes exactly half the original difference ($A - B$). Find A .

- A) 25
- B) 35
- C) 30
- D) 40

14

FYQ · MCQ

In a two-candidate election with base votes split 65:35 (X leading), each candidate independently chooses a campaign intensity level i (for X) and j (for Y), where $i, j \in \{0, 1, 2\}$. The rules: X's base loses $8j\%$ (switching to Y) and $4i\%$ (abstaining) of its original 65 units; Y's base loses $8i\%$ (switching to X) and $4j\%$ (abstaining) of its original 35 units — all percentages measured on the ORIGINAL base values. X chooses i FIRST to maximize the final margin; Y then chooses j (after observing i) to MINIMIZE the final margin. Find the margin X can guarantee.

- A) 12
- B) 18
- C) 21
- D) 15

15

FYQ · MCQ

An index starts at 125. It rises by 56.25% in every year that is a multiple of 3, and falls by 20% in every other year. Find the index value after exactly 50 years.

- A) 80
- B) 100
- C) 60
- D) 125

16

FYQ · MCQ

In an economy, income shares of the Top 10%, Middle 80%, and Bottom 10% earners always sum to exactly 100%. A certain index Z is affected as follows: every 1-percentage-point rise in the Top 10%'s share causes Z to fall by 0.2%; every 1-point rise in the Middle 80%'s share causes Z to rise by 0.1%; every 1-point rise in the Bottom 10%'s share causes Z to rise by 0.5% (all effects independent and linear). In a

given year, the Top 10%'s share rises by 3 points and the Bottom 10%'s share falls by 5 points. Find the net percentage change in Z.

- A) -2%
- B) -3.5%
- C) -1.9%
- D) -2.9%

17

FYQ · MCQ

Country A exports exclusively to three countries: B, C, and D. Country B's total imports are 1000 units, of which A's share is 18%. Country C's total imports are 1500 units, of which A's share is 20%. Additionally, A's exports to D are known to equal exactly 40% of A's OWN total exports (a self-referential condition). Find A's percentage share of Country D's total imports, given that D's total imports are 800 units.

- A) 35%
- B) 45%
- C) 40%
- D) 30%

18

FYQ · MCQ

A stadium has capacity 5000. In Scenario 1, Gate A admits visitors until occupancy reaches 60% of capacity; then simultaneously, $x\%$ of those inside leave via Gate C, and e visitors enter via Gate B, bringing final occupancy to 70%. In Scenario 2, Gate A instead admits visitors until occupancy reaches only 50% of capacity; then the SAME $x\%$ leave via Gate C, and the SAME e visitors enter via Gate B, bringing final occupancy to 64%. Find e .

- A) 1500
- B) 1700
- C) 1800
- D) 1600

19

FYQ · MCQ

A three-digit number N , when increased by 20%, becomes a number whose digits are exactly the digits of N written in reverse order. Find N .

- A) 495
- B) 594
- C) 459
- D) 549

20

FYQ · MCQ

Two positive integers X and Y satisfy $X+Y=1200$, with X a multiple of 40. X is increased by 25%, and Y is decreased by $L\%$ (where L is a positive multiple of 5, $L \leq 70$), resulting in a combined total of exactly 1260 (5% more than 1200). This equation alone admits FIVE valid (X,L) candidate pairs. Given TWO additional

clues — (1) Y lies strictly between 300 and 500, and (2) L is NOT a multiple of 25 — find the unique value of X.

- A) 900
- B) 700
- C) 850
- D) 800

21

FYQ · MCQ

A country's population is split into three age brackets — 0-18, 18-60, and 60+ — with shares always summing to exactly 100%. Initially the shares are 35%, 50%, and 15% respectively. Over a decade, the 0-18 share falls by 5 percentage points, and the 60+ share rises by 9 percentage points. Find the percentage CHANGE (not percentage-point change) in the 18-60 share.

- A) -10%
- B) -6%
- C) -8%
- D) -12%

22

FYQ · MCQ

If A is increased by 25%, it becomes B. If B is decreased by 20%, it becomes C. If C is increased by 25%, it becomes 300. Further, the difference between B and A is exactly 60. Find A.

- A) 240
- B) 300
- C) 192
- D) 288

23

FYQ · MCQ

A town's population changes over three consecutive decades: it rises by $r_1\%$ in decade 1, falls by $(r_1-10)\%$ in decade 2, and rises by $(r_1+5)\%$ in decade 3. If the net change over all three decades is exactly +35%, find r_1 , where r_1 is a positive integer (assume $r_1 > 10$ so that decade 2's rate is a valid positive decrease).

- A) 20
- B) 15
- C) 25
- D) 30

24

FYQ · MCQ

A rectangle's length is measured with an excess error of $e_1\%$, and its width with an excess error of $(e_1+2)\%$. This results in a calculated area that is 23.2% MORE than the actual area. Find e_1 . If the rectangle's actual length:width ratio is 3:2, find the resulting percentage error in the calculated PERIMETER.

- A)** $e_1 = 8$, perimeter error = 10.8%
C) $e_1 = 12$, perimeter error = 11.5%

- B)** $e_1 = 10$, perimeter error = 9.6%
D) $e_1 = 10$, perimeter error = 10.8%

25

FYQ · MCQ

x is a multiple of 10. When x is increased by $x\%$ of itself, the result is a perfect square. Find the unique value of x (among multiples of 10 up to 1000).

- A)** 40
C) 80

- B)** 90
D) 60

26

FYQ · MCQ

A quantity undergoes two successive increases of $p\%$ and $(p+10)\%$. Had the order of the two increases been reversed, the final value would have remained unchanged. However, the average of the intermediate values obtained in the two possible orders exceeds the original quantity by exactly 25%, while the common final value exceeds the original quantity by exactly 56%. Find p .

- A)** 15
C) 25

- B)** 20
D) 30

27

FYQ · MCQ

A city's population consists of males and females, with the total population initially equal to 12,000. During the following year, the male population increases by 10% and then receives an additional net immigration of 600 people, while the female population increases by 10% and then loses 200 people through net emigration. After these changes, the ratio of males to females becomes exactly 9:8. Find the original number of males.

- A)** 5,000
C) 6,000

- B)** 7,200
D) 6,500

28

FYQ · MCQ

A quantity is subjected to three successive reductions. The first reduction is 20%, the second reduction is $d\%$, and the third reduction is $(d + 10)\%$. If the overall reduction is exactly 55.2%, find d .

- A)** 25%
C) 30%

- B)** 20%
D) 15%

29**FYQ · MCQ**

A committee has 500 seats, of which 40% are reserved for Category R. Within Category R, 25% of the seats are further reserved for Category S. Category S fills 60% of its allotted seats. Of the remaining unfilled S seats, 75% are transferred to the general pool, while the rest remain reserved. Separately, 40% of the Category R seats outside the S quota remain unfilled, and half of these unfilled seats are also transferred to the general pool. Find the final number of seats in the general pool.

A) 300**B)** 330**C)** 360**D)** 345**30****FYQ · MCQ**

A sequence is defined by $a_1=100$, $a_2=110$, and for $n \geq 3$: $a(n) = a(n-1) \times (1 + a(n-2)/100)$. Find a_4 .

A) 420**B)** 462**C)** 440**D)** 484**31****FYQ · MCQ**

A tank has a capacity of 20,000 litres and initially contains water equal to 40% of its capacity. At the beginning of every hour, a pump adds water equal to 20% of the tank's capacity. Immediately afterwards, one-fourth of the water present in the tank is drained. Finally, before the next hour begins, an automatic control system adjusts the water level by adding or removing an amount equal to 20% of the difference between the current water level and half the tank's capacity, whichever is required to move the water level towards half-capacity. After how many complete hours will the amount of water in the tank first exceed 10,500 litres?

A) 4 hours**B)** 3 hours**C)** 5 hours**D)** 6 hours**32****FYQ · MCQ**

A trader evaluates the movement of a currency over two months. The first month's percentage gain exceeds the second month's percentage loss by 8 percentage points. Believing that gains and losses combine arithmetically, the trader expects the exchange rate to rise by 8% overall. The realised increase, however, is only 5.6%. Assuming the exchange rate changes only once in each month, find the first month's percentage appreciation.

A) 15%**B)** 18%**C)** 25%**D)** 20%

exactly +5%, find x.

- A) 0%
- B) 10%
- C) 5%
- D) -5%

38

FYQ · MCQ

A battery starts at an unknown charge level $L\%$. Each hour for 2 hours, it drains 15% of its CURRENT charge (multiplicative loss). After these 2 hours, it is charged by a FIXED 20 percentage-points. If the final charge level is exactly 77.8%, find L .

- A) 70%
- B) 80%
- C) 75%
- D) 85%

39

FYQ · MCQ

Country A exports to four destinations: B, C, D, and E. A fixed 100 units go to E. Of the REMAINING exports, 30% go to B, 45% go to C, and 25% go to D. Country B's total imports are 900 units, of which A's share is 20%. Find Country A's total exports T , and then find A's percentage share of Country D's imports (given D's total imports are 600 units).

- A) $T = 700$, D-share = 25%
- B) $T = 750$, D-share = 20%
- C) $T = 680$, D-share = 22%
- D) $T = 720$, D-share = 28%

40

FYQ · MCQ

Three exam papers have maximum marks 200, 300, and 500 (total 1000). A student needs 45% of the TOTAL marks (450) to pass. If the student scores 60% in Paper 1 and 50% in Paper 2, find the minimum percentage needed in Paper 3 to pass overall.

- A) 30%
- B) 40%
- C) 32%
- D) 36%

41

FYQ · MCQ

A factory processes 1000 kg of raw material. Stage 1 (cutting) wastes 12% of the material. Stage 2 (shaping) wastes 8% of what remains after Stage 1. Stage 3 (polishing) wastes a FIXED w kg. If the final usable output is exactly 780 kg, find w .

- A) 25 kg
- B) 30 kg

C) 29.6 kg

D) 32.4 kg

42

FYQ · MCQ

A two-digit positive integer N exceeds the sum of its digits by exactly $k\%$, where k is an integer. Reversing the digits decreases the number by 9, and the tens digit is twice the units digit. Find the value of k . Further, determine how many other two-digit positive integers have the same value of k .

A) $k = 500\%$, 2 other numbers

B) $k = 600\%$, 3 other numbers

C) $k = 700\%$, 4 other numbers

D) $k = 600\%$, 2 other numbers

43

FYQ · MCQ

Country A exports to four destinations: B, C, D, and E. A fixed 100 units go to E, and the remaining $(T-100)$ units are split 30:45:25 among B, C, and D respectively. Country B's imports are 900 units, of which A's share is 20%. Find A's total exports T , and then find A's percentage share of Country D's imports (D's total imports = 600 units).

A) $T = 700$, D-share = 25%

B) $T = 650$, D-share = 20%

C) $T = 720$, D-share = 30%

D) $T = 680$, D-share = 22%

44

FYQ · MCQ

A company has four pay-bands with employee COUNTS in the ratio 4:3:2:1, and average pay per employee in each band in the ratio 1:2:3:4 (Band 1 has the most employees but the lowest average pay). Find Band 1's total wage bill as a percentage of the company's total wage bill.

A) 15%

B) 25%

C) 30%

D) 20%

45

FYQ · MCQ

A township developer earmarks 25% of the total land for infrastructure, reserves 20% of the balance for utilities, and allocates to the clubhouse an area equal to the difference between 15% of the original tract and 5% of the land remaining after the second stage; if only 75% of the residential land is sold and the sold portion measures 91.8 hectares, find the original area of the township.

A) 225 hectares

B) 240 hectares

C) 255 hectares

D) 270 hectares

46

FYQ · MCQ

An election has 2,000 ballots, of which 5% are found invalid after the first count. Among the remaining valid votes, Candidates A, B and C receive votes in the ratio 18 : 15 : 7. Candidate C is eliminated. Of C's votes, 25% are exhausted (cannot be transferred), while the remaining votes are split between A and B in the ratio 2 : 5. By what percentage does Candidate A's final vote tally exceed Candidate B's final vote tally?

- A) 2%
- C) 5%

- B) 4%
- D) 6%

47

FYQ · MCQ

Two investments, A and B, both start at 1000. Investment A grows LINEARLY — each period, it gains a fixed 20% of its ORIGINAL value (1000). Investment B grows MULTIPLICATIVELY — each period, it gains 20% of its CURRENT value. Find the first period in which difference of investment of A and B exceed 1,00,000.

- A) 26th period
- C) 25th period

- B) 24th period
- D) 27th period

48

FYQ · MCQ

A survey is sent to 2000 people, with a 45% response rate. Among respondents, 60% favor Policy X. It's estimated that non-respondents favor Policy X at a rate 20 percentage points LOWER than respondents. Estimate the overall favor-X rate across ALL 2000 people (respondents plus estimated non-respondents).

- A) 45%
- C) 52%

- B) 50%
- D) 49%

49

FYQ · MCQ

A reservoir has capacity 10,000 units and starts at 80% full. Each month, it loses 25% of its CURRENT volume to evaporation, and then gains a FIXED 400 units from rainfall. Find the reservoir's level, as a percentage of capacity, after 3 months.

- A) 40%
- C) 43%

- B) 45%
- D) 48%

50

FYQ · MCQ

Two positive integers X and Y satisfy $X+Y=1800$, with X a multiple of 50. X is increased by 30%, and Y is

decreased by $M\%$ (a positive multiple of 5, $M \leq 80$), resulting in a combined total of exactly 1890 (5% more than 1800). This equation alone admits SIX valid (X, M) candidate pairs. Given THREE additional clues — (1) Y is greater than 700, (2) M is an even number, and (3) the absolute decrease amount on Y is a multiple of 75 — find the unique value of X .

A) 900

B) 1050

C) 1200

D) 1300

02 Answer Key

Quick-reference correct options for all 50 FYQs

Q01 C	Q02 B	Q03 A	Q04 D	Q05 C
Q06 A	Q07 A	Q08 C	Q09 B	Q10 D
Q11 A	Q12 D	Q13 C	Q14 B	Q15 A
Q16 D	Q17 C	Q18 B	Q19 A	Q20 D
Q21 C	Q22 A	Q23 A	Q24 D	Q25 C
Q26 B	Q27 C	Q28 B	Q29 D	Q30 B
Q31 A	Q32 D	Q33 C	Q34 B	Q35 A
Q36 D	Q37 C	Q38 B	Q39 A	Q40 D
Q41 C	Q42 B	Q43 A	Q44 D	Q45 C
Q46 B	Q47 A	Q48 D	Q49 C	Q50 B

03 Solutions

Stepwise solutions for every FYQ

01

FYQ C

A vessel contains V litres of a solution that is 50% acid. First, 20% of the solution is drained and replaced with water. Then, 30% of the resulting mixture is drained and replaced with pure acid. Finally, 25% of this new mixture is drained and replaced with water. If the final amount of acid in the vessel is exactly 130.5 litres, find V .

STEPWISE SOLUTION

Step 1: Find the initial amount of acid

Initial acid = 50% of $V = 0.5V$

Step 2: Apply Stage 1 (drain 20%, replace with water)

Acid remaining = $0.5V \times 0.8 = 0.4V$

Step 3: Apply Stage 2 (drain 30%, replace with pure acid)

Acid removed = $0.4V \times 0.3 = 0.12V$

Acid remaining after removal = $0.4V - 0.12V = 0.28V$

Pure acid added = $0.3V$

Acid after Stage 2 = $0.28V + 0.3V = 0.58V$

Step 4: Apply Stage 3 (drain 25%, replace with water)

Acid remaining = $0.58V \times 0.75 = 0.435V$

Step 5: Set up and solve the equation

$0.435V = 130.5$

$V = 130.5 / 0.435 = 300$

Answer: $V = 300$ litres

02

FYQ E

A class has 420 students, consisting only of boys and girls. In the first scenario, 20% of the boys and 25% of the girls leave the class. Subsequently, 12 new boys and 15 new girls join. After these changes, the number of girls is exactly 25% more than the number of boys. In a second scenario, 30% of the original boys and 15% of the original girls leave instead, followed by the same 12 boys and 15 girls joining. In this case, the number of girls exceeds the number of boys by 81. Find the original number of boys.

STEPWISE SOLUTION

Step 1: Define variables

Let the original number of boys be x , so the original number of girls is $420 - x$.

Step 2: Apply the first scenario's changes

Boys after leaving/joining = $0.8x + 12$

Girls after leaving/joining = $0.75(420 - x) + 15$

Step 3: Set up the equation from the "25% more" condition

$0.75(420 - x) + 15 = 1.25(0.8x + 12)$

Step 4: Simplify and solve

$$315 - 0.75x + 15 = x + 15$$

$$315 = 1.75x$$

$$x = 180$$

Step 5: Verify using the second scenario

$$\text{Boys} = 0.7 \times 180 + 12 = 138$$

$$\text{Girls} = 0.85 \times 240 + 15 = 219$$

$$\text{Difference} = 219 - 138 = 81 \checkmark \text{ matches the given condition}$$

Answer: 180

03

FYQ A

A number is increased by $p\%$, then decreased by $q\%$, where p and q are positive integers with $p+q=50$. If the net result is exactly a 4% increase over the original number, find p .

STEPWISE SOLUTION**Step 1: Express q in terms of p**

$$q = 50 - p$$

Step 2: Set up the net multiplier equation

$$(1 + p/100)(1 - q/100) = 1.04$$

$$(100+p)(100-q)/10000 = 1.04$$

$$(100+p)(100-(50-p)) = 10,400$$

$$(100+p)(50+p) = 10,400$$

Step 3: Expand and form a quadratic

$$5000 + 150p + p^2 = 10,400$$

$$p^2 + 150p - 5,400 = 0$$

Step 4: Solve using the quadratic formula

$$\text{Discriminant} = 150^2 + 4(5,400) = 22,500 + 21,600 = 44,100 = 210^2$$

$$p = (-150 + 210) / 2 = 30 \text{ (rejecting the negative root)}$$

Answer: $p = 30$

04

FYQ D

A person's gross monthly salary is S . First, 20% of the salary is deducted as tax. From the amount remaining, a fixed bonus of ₹1,500 is added. Then 15% of this new amount is deducted as insurance. Finally, ₹1,200 is deducted as EMI. The final take-home salary is ₹8,235. However, the person later receives a revised salary structure in which the tax deduction becomes 25% of the gross salary, the bonus becomes ₹2,500, the insurance deduction becomes 10% of the amount after adding the bonus, and the EMI remains unchanged. Under the revised structure, the take-home salary would be ₹9,150. Find S .

STEPWISE SOLUTION

Step 1: Apply the 20% tax deduction

Amount after tax = $0.8S$

Step 2: Add the ₹1,500 bonus

Amount after bonus = $0.8S + 1,500$

Step 3: Deduct 15% insurance and then the ₹1,200 EMI, setting the result equal to the take-home

$$0.85(0.8S + 1,500) - 1,200 = 8,235$$

Step 4: Simplify and solve

$$0.68S + 1,275 - 1,200 = 8,235$$

$$0.68S + 75 = 8,235$$

$$0.68S = 8,160$$

$$S = 12,000$$

Step 5: Verify using the revised structure

Tax: $0.75 \times 12,000 = 9,000$

Bonus: $9,000 + 2,500 = 11,500$

Insurance: $0.90 \times 11,500 = 10,350$

EMI: $10,350 - 1,200 = 9,150$ ✓ matches the given revised take-home

Answer: $S = ₹12,000$

05

FYQ C

Three friends' incomes are in the ratio 5:6:9. They contribute 10%, 20%, and $r\%$ of their respective incomes toward a shared bill, and this combined contribution equals 22% of their total income. Find r . If their incomes had instead been in the REVERSED ratio 9:6:5 (with the same three contribution rates 10%, 20%, and $r\%$ now applied in that new order), what would the combined contribution be as a percentage of total income?

STEPWISE SOLUTION**Step 1: Set up the weighted-average equation for r**

$$(5 \times 10 + 6 \times 20 + 9 \times r) / (5 + 6 + 9) = 22$$

Step 2: Simplify and solve for r

$$(50 + 120 + 9r) / 20 = 22$$

$$170 + 9r = 440$$

$$9r = 270$$

$$r = 30$$

Step 3: Apply the same rates to the reversed ratio (9,6,5)

$$\text{Contribution} = 9 \times 10 + 6 \times 20 + 5 \times 30$$

$$= 90 + 120 + 150 = 360$$

Step 4: Express as a percentage of total income

$$\text{New\%} = 360 / 20 = 18\%$$

Answer: $r = 30$, new% = 18%

06**FYQ A**

Two factories, A and B, initially produce the same total output. Initially, the output per worker in Factory A is 75% of the output per worker in Factory B, and Factory A has 240 more workers than Factory B. During the following year, the output per worker in Factory A increases by 25%, after which 20% of its workers leave, and the productivity of each remaining worker increases by a further 8%. In Factory B, the output per worker decreases by 10%, the workforce increases by 25%, and then 20% of the resulting workforce leaves. After all these changes, Factory A's total output is exactly 20% greater than Factory B's total output. Find the original number of workers in Factory A.

STEPWISE SOLUTION**Step 1: Set up variables**

Let Factory B's original workers = x , so Factory A's original workers = $x + 240$.

Let Factory B's output per worker = P , so Factory A's is $0.75P$.

Step 2: Use the equal-initial-output condition

$$(x + 240)(0.75P) = xP$$

$$0.75(x + 240) = x$$

$$0.75x + 180 = x$$

$$0.25x = 180$$

$$x = 720$$

Step 3: Find Factory A's original workers

$$\text{Factory A's original workers} = 720 + 240 = 960$$

Step 4: Verify using the final-output condition

Factory A's productivity multiplier = $1.25 \times 1.08 = 1.35$; workforce falls to 80%, so output multiplier = $1.35 \times 0.8 = 1.08$

Factory B's productivity multiplier = 0.9; workforce rises 25% then falls 20%, returning to original size, so output multiplier = 0.9

Ratio = $1.08 : 0.9 = 1.2 : 1$ ✓ matches the given 20% condition

Answer: 960 workers**07****FYQ A**

In a survey, 70% of respondents use Product A, 80% use Product B, and 60% use Product C. Find the difference between the MAXIMUM possible percentage using all three products and the MINIMUM possible percentage using all three products.

STEPWISE SOLUTION**Step 1: Find the minimum percentage using all three (inclusion-exclusion bound)**

$$\text{Minimum} = 70 + 80 + 60 - 2(100) = 210 - 200 = 10\%$$

Step 2: Find the maximum percentage using all three (subset bound)

$$\text{Maximum} = \min(70, 80, 60) = 60\%$$

Step 3: Find the difference

$$\text{Difference} = 60 - 10 = 50$$

Answer: 50

08**FYQ C**

Two positive integers x and y satisfy the following conditions: when x is decreased by $y\%$, the result is 48, and when y is increased by $x\%$, the result is 72. If x is increased by 20% while y is decreased by 20%, the resulting difference between the two numbers is 64. Find x .

STEPWISE SOLUTION**Step 1: Set up the two equations**

$$x - (xy/100) = 48 \dots(1)$$

$$y + (xy/100) = 72 \dots(2)$$

Step 2: Add the equations

$$x + y = 120, \text{ so } y = 120 - x$$

Step 3: Substitute into equation (1)

$$x - x(120 - x)/100 = 48$$

$$100x - 120x + x^2 = 4800$$

$$x^2 - 20x - 4800 = 0$$

Step 4: Factor and solve

$$(x - 80)(x + 60) = 0$$

Since x is positive, $x = 80$, so $y = 40$

Step 5: Verify using the third condition

$$x \text{ increased by } 20\% = 80 \times 1.20 = 96$$

$$y \text{ decreased by } 20\% = 40 \times 0.80 = 32$$

$$\text{Difference} = 96 - 32 = 64 \checkmark \text{ matches the given condition}$$

Answer: $x = 80$

09**FYQ B**

A town has a population of 3,000 at the beginning of Year 1. At the end of each year, its population increases by 20%, after which 100 people migrate out. However, at the end of every third year, an additional 200 people migrate out after the regular migration. Find the first year at the end of which the population exceeds 12,000.

STEPWISE SOLUTION**Step 1: Derive the 3-year cycle formula**

$$\text{After year 1: } 1.2P - 100$$

$$\text{After year 2: } 1.2(1.2P - 100) - 100 = 1.44P - 220$$

$$\text{After year 3 (with extra 200 out): } 1.2(1.44P - 220) - 100 - 200 = 1.728P - 564$$

Step 2: Apply the cycle for the first 3 years ($P = 3,000$)

$$1.728 \times 3,000 - 564 = 5,184 - 564 = 4,620$$

Step 3: Apply the cycle for years 4–6 ($P = 4,620$)

$$1.728 \times 4,620 - 564 = 7,983.36 - 564 = 7,419.36$$

Step 4: Compute Year 7

$$7,419.36 \times 1.2 - 100 = 8,903.232 - 100 = 8,803.232$$

Step 5: Compute Year 8

$$8,803.232 \times 1.2 - 100 = 10,563.8784 - 100 = 10,463.8784 \text{ (still below 12,000)}$$

Step 6: Compute Year 9 (with the extra 200 out)

$$10,463.8784 \times 1.2 - 100 - 200 = 12,556.65408 - 300 = 12,256.65408 \text{ (exceeds 12,000)}$$

Answer: 9th year

10

FYQ D

Country A exports a total of T units to Countries B, C, D and E. A fixed 120 units are exported to D, while the remaining exports are divided among B, C and E in the ratio 2:3:1. Country B imports 1,500 units in total, of which 16% are supplied by Country A. Find the total exports T of Country A.

STEPWISE SOLUTION

Step 1: Find A's exports to B

$$\text{Exports to B} = 16\% \text{ of } 1,500 = 240 \text{ units}$$

Step 2: Use the 2:3:1 ratio to find total exports to B, C and E

B's share is 2 out of $(2+3+1) = 6$ parts, i.e., $1/3$ of the total to B, C, E

$$\text{So exports to B, C and E} = 240 \times 3 = 720 \text{ units}$$

Step 3: Add the fixed exports to D

$$\text{Total exports } T = 720 + 120$$

Step 4: Compute T

$$T = 840 \text{ units}$$

Answer: 840 units

11

FYQ A

A quantity undergoes three successive percentage increases, $a\%$, $b\%$, and $c\%$, where $a+b+c=90$ (all positive). What is the maximum possible overall percentage increase in the quantity?

STEPWISE SOLUTION

Step 1: Set up the product to be maximized

$$\text{Maximize } (1+a/100)(1+b/100)(1+c/100) \text{ subject to } a+b+c=90$$

Step 2: Apply AM-GM for three terms

The product of three positive factors with a fixed sum is maximized when all three are equal.

$$\text{So } a = b = c = 30$$

Step 3: Compute the maximum multiplier

$$\text{Multiplier} = (1.30)^3 = 2.197$$

Step 4: Convert to percentage increase

$$\text{Overall increase} = 119.7\%$$

Answer: 119.7%

12FYQ **D**

Let N be a positive integer such that $1.25N$ lies strictly between 150 and 200, $0.60N$ lies strictly between 80 and 130, N is a multiple of 6, and N is NOT a multiple of 5. How many values of N satisfy all four conditions?

STEPWISE SOLUTION**Step 1: Solve the first range condition**

$$1.25N \in (150, 200) \rightarrow N \in (120, 160)$$

Step 2: Solve the second range condition

$$0.60N \in (80, 130) \rightarrow N \in (133.33, 216.67)$$

Step 3: Find the intersection of both ranges

$$N \in (133.33, 160) \rightarrow \text{integer values } 134 \text{ to } 159$$

Step 4: List multiples of 6 in this range

138, 144, 150, 156

Step 5: Exclude multiples of 5

150 is a multiple of 5, so exclude it.

Remaining values: 138, 144, 156

Answer: 3 values

13FYQ **C**

Two positive numbers A and B satisfy $A \times B = 600$ (their *PRODUCT*, not their sum), with $A > B$. If A increases by 10% and B increases by 40%, the new difference ($A_{\text{new}} - B_{\text{new}}$) becomes exactly half the original difference ($A - B$). Find A .

STEPWISE SOLUTION**Step 1: Set up the difference condition**

$$1.10A - 1.40B = 0.5(A - B)$$

Step 2: Simplify

$$1.10A - 1.40B = 0.5A - 0.5B$$

$$0.60A = 0.90B$$

$$A = 1.5B$$

Step 3: Substitute into the product condition $AB = 600$

$$(1.5B)(B) = 600$$

$$1.5B^2 = 600$$

$$B^2 = 400$$

$$B = 20$$

Step 4: Find A

$$A = 1.5 \times 20 = 30$$

Answer: $A = 30$

14**FYQ E**

In a two-candidate election with base votes split 65:35 (X leading), each candidate independently chooses a campaign intensity level i (for X) and j (for Y), where $i, j \in \{0, 1, 2\}$. The rules: X's base loses $8j\%$ (switching to Y) and $4i\%$ (abstaining) of its original 65 units; Y's base loses $8i\%$ (switching to X) and $4j\%$ (abstaining) of its original 35 units — all percentages measured on the ORIGINAL base values. X chooses i FIRST to maximize the final margin; Y then chooses j (after observing i) to MINIMIZE the final margin. Find the margin X can guarantee.

STEPWISE SOLUTION**Step 1: Set up the new-vote formulas**

$$\text{New X} = 65 - 8j\%(65) - 4i\%(65) + 8i\%(35)$$

$$\text{New Y} = 35 - 8i\%(35) - 4j\%(35) + 8j\%(65)$$

Step 2: Evaluate the margin when X plays its strongest strategy, $i=2$

$$\text{Margin} = 36 - 9j$$

$$\text{For } j=0, 1, 2: \text{ margins} = 36, 27, 18$$

Step 3: Determine Y's best response

Y minimizes the margin, so Y picks $j=2$, giving margin = 18

Step 4: Check that $i=2$ is indeed X's best choice

For $i=0$ and $i=1$, checking Y's best responses gives worst-case margins of 12 and 15 respectively — both lower than 18.

So X's optimal choice is $i=2$.

Answer: X can guarantee a margin of 18

15**FYQ A**

An index starts at 125. It rises by 56.25% in every year that is a multiple of 3, and falls by 20% in every other year. Find the index value after exactly 50 years.

STEPWISE SOLUTION**Step 1: Check the invariant over any 3-year block**

$$1.5625 \times 0.8 \times 0.8 = 1.5625 \times 0.64 = 1.00$$

So every complete 3-year block leaves the index unchanged.

Step 2: Break 50 years into complete blocks and leftover years

$$50 = 16 \text{ complete blocks (48 years)} + 2 \text{ leftover years}$$

Step 3: Compute the index after 48 years

Index remains 125 (unchanged after complete blocks)

Step 4: Compute Year 49 (not a multiple of 3, so a "fall" year)

$$125 \times 0.8 = 100$$

Step 5: Compute Year 50 (not a multiple of 3, so a "fall" year)

$$100 \times 0.8 = 80$$

Answer: 80

16FYQ **D**

In an economy, income shares of the Top 10%, Middle 80%, and Bottom 10% earners always sum to exactly 100%. A certain index Z is affected as follows: every 1-percentage-point rise in the Top 10%'s share causes Z to fall by 0.2%; every 1-point rise in the Middle 80%'s share causes Z to rise by 0.1%; every 1-point rise in the Bottom 10%'s share causes Z to rise by 0.5% (all effects independent and linear). In a given year, the Top 10%'s share rises by 3 points and the Bottom 10%'s share falls by 5 points. Find the net percentage change in Z .

STEPWISE SOLUTION**Step 1: Use the constraint that all three shares sum to 100%**

$$\Delta\text{Top} + \Delta\text{Middle} + \Delta\text{Bottom} = 0$$

Step 2: Solve for the Middle group's change

$$3 + \Delta\text{Middle} + (-5) = 0$$

$$\Delta\text{Middle} = 2$$

Step 3: Apply the sensitivity coefficients

$$\text{Net change in } Z = (-0.2 \times 3) + (0.1 \times 2) + (0.5 \times -5)$$

Step 4: Compute

$$= -0.6 + 0.2 - 2.5 = -2.9\%$$

Answer: -2.9%**17**FYQ **C**

Country A exports exclusively to three countries: B, C, and D. Country B's total imports are 1000 units, of which A's share is 18%. Country C's total imports are 1500 units, of which A's share is 20%. Additionally, A's exports to D are known to equal exactly 40% of A's OWN total exports (a self-referential condition). Find A's percentage share of Country D's total imports, given that D's total imports are 800 units.

STEPWISE SOLUTION**Step 1: Find A's exports to B and C**

$$\text{Exports to B} = 0.18 \times 1000 = 180$$

$$\text{Exports to C} = 0.20 \times 1500 = 300$$

Step 2: Set up the self-referential total-exports equation

$$\text{Let } T = \text{A's total exports. Exports to D} = 0.40T$$

$$T = 180 + 300 + 0.40T$$

Step 3: Solve for T

$$0.60T = 480$$

$$T = 800$$

Step 4: Find A's exports to D

$$\text{Exports to D} = 0.40 \times 800 = 320$$

Step 5: Find A's percentage share of D's imports

$$\text{Share} = 320 / 800 \times 100 = 40\%$$

Answer: 40%

18**FYQ E**

A stadium has capacity 5000. In Scenario 1, Gate A admits visitors until occupancy reaches 60% of capacity; then simultaneously, $x\%$ of those inside leave via Gate C, and e visitors enter via Gate B, bringing final occupancy to 70%. In Scenario 2, Gate A instead admits visitors until occupancy reaches only 50% of capacity; then the SAME $x\%$ leave via Gate C, and the SAME e visitors enter via Gate B, bringing final occupancy to 64%. Find e .

STEPWISE SOLUTION**Step 1: Set up the equation for Scenario 1**

$$3000(1 - x/100) + e = 3500$$

Step 2: Set up the equation for Scenario 2

$$2500(1 - x/100) + e = 3200$$

Step 3: Subtract Scenario 2 from Scenario 1 to eliminate e

$$500(1 - x/100) = 300$$

$$1 - x/100 = 0.6$$

$$x = 40$$

Step 4: Substitute x back into the Scenario 1 equation

$$3000(0.6) + e = 3500$$

$$1800 + e = 3500$$

$$e = 1700$$

Answer: $e = 1700$ **19****FYQ A**

A three-digit number N , when increased by 20%, becomes a number whose digits are exactly the digits of N written in reverse order. Find N .

STEPWISE SOLUTION**Step 1: Set up the digit representation**

Let $N = 100h + 10t + u$, and the reversed number $R = 100u + 10t + h$

Step 2: Set up the reversal condition

$$R = 1.2N$$

$$100u + 10t + h = 1.2(100h + 10t + u)$$

Step 3: Clear the decimal by multiplying through by 5

$$500u + 50t + 5h = 600h + 60t + 6u$$

$$494u = 595h + 10t$$

Step 4: Test $h = 4$

$$494u = 2380 + 10t$$

Step 5: Test $u = 5$

$$2470 = 2380 + 10t$$

$$t = 9$$

Step 6: Verify

$N = 495$; $495 \times 1.2 = 594$, which is 495 with digits reversed ✓

Answer: N = 495

20

FYQ D

Two positive integers X and Y satisfy $X+Y=1200$, with X a multiple of 40. X is increased by 25%, and Y is decreased by $L\%$ (where L is a positive multiple of 5, $L \leq 70$), resulting in a combined total of exactly 1260 (5% more than 1200). This equation alone admits FIVE valid (X,L) candidate pairs. Given TWO additional clues — (1) Y lies strictly between 300 and 500, and (2) L is NOT a multiple of 25 — find the unique value of X .

STEPWISE SOLUTION

Step 1: Solve the core equation for all valid (X,L) pairs

$$1.25X + (1200 - X)(1 - L/100) = 1260, \text{ with } X \text{ a multiple of 40 and } L \text{ a positive multiple of 5 } (\leq 70)$$

This yields five candidates: (400,5), (600,15), (720,25), (800,35), (880,50)

Step 2: Apply Clue 1 ($300 < Y < 500$, i.e., $700 < X < 900$)

Candidates satisfying this: $X = 720, 800, 880$

Step 3: Apply Clue 2 (L not a multiple of 25) to the full candidate set

$L = 25$ and $L = 50$ are multiples of 25, so eliminate $X = 720$ and $X = 880$

Candidates satisfying this: $X = 400, 600, 800$

Step 4: Find the intersection of both clue sets

$$\{720, 800, 880\} \cap \{400, 600, 800\} = \{800\}$$

Answer: X = 800

21

FYQ C

A country's population is split into three age brackets — 0-18, 18-60, and 60+ — with shares always summing to exactly 100%. Initially the shares are 35%, 50%, and 15% respectively. Over a decade, the 0-18 share falls by 5 percentage points, and the 60+ share rises by 9 percentage points. Find the percentage CHANGE (not percentage-point change) in the 18-60 share.

STEPWISE SOLUTION

Step 1: Use the constraint that all three shares sum to 100%

$$\Delta(0-18) + \Delta(18-60) + \Delta(60+) = 0$$

Step 2: Solve for the forced change in the 18-60 share

$$-5 + x + 9 = 0$$

$$x = -4 \text{ (points)}$$

Step 3: Convert the point-change into a percentage change relative to its own original value

The 18-60 bracket started at 50%.

$$\text{Percentage change} = -4 / 50 \times 100 = -8\%$$

Answer: -8%

22FYQ **A**

If A is increased by 25%, it becomes B . If B is decreased by 20%, it becomes C . If C is increased by 25%, it becomes 300. Further, the difference between B and A is exactly 60. Find A .

STEPWISE SOLUTION**Step 1: Use the $B - A$ condition**

$$B = 1.25A$$

$$B - A = 60, \text{ so } 1.25A - A = 60$$

Step 2: Solve for A

$$0.25A = 60$$

$$A = 240$$

Step 3: Verify using the remaining conditions

$$B = 1.25 \times 240 = 300$$

$$C = 80\% \text{ of } 300 = 240$$

$$\text{Increasing } C \text{ by } 25\%: 240 \times 1.25 = 300 \checkmark \text{ matches the given final condition}$$

Answer: $A = 240$ **23**FYQ **A**

A town's population changes over three consecutive decades: it rises by $r1\%$ in decade 1, falls by $(r1-10)\%$ in decade 2, and rises by $(r1+5)\%$ in decade 3. If the net change over all three decades is exactly +35%, find $r1$, where $r1$ is a positive integer (assume $r1 > 10$ so that decade 2's rate is a valid positive decrease).

STEPWISE SOLUTION**Step 1: Test $r1 = 20$**

Decade 1 rate: +20%

Decade 2 rate: $r1 - 10 = 10\%$ decrease

Decade 3 rate: $r1 + 5 = 25\%$ increase

Step 2: Compute the net multiplier

$$\text{Multiplier} = 1.20 \times 0.90 \times 1.25$$

Step 3: Calculate

$$= 1.20 \times 0.90 \times 1.25 = 1.35$$

Step 4: Verify against the given condition

1.35 corresponds to a +35% net change, matching the given condition $\checkmark$

Answer: $r1 = 20$ **24**FYQ **D**

A rectangle's length is measured with an excess error of $e1\%$, and its width with an excess error of $(e1+2)\%$. This results in a calculated area that is 23.2% MORE than the actual area. Find $e1$. If the rectangle's actual length:width

ratio is 3:2, find the resulting percentage error in the calculated PERIMETER.

STEPWISE SOLUTION

Step 1: Set up the area-error equation

$$(1 + e_1/100)(1 + (e_1+2)/100) = 1.232$$

Step 2: Test $e_1 = 10$

$$(1.10)(1.12) = 1.232 \checkmark$$

Step 3: Set up the actual and measured dimensions using the 3:2 ratio

Let actual length = $3k$, actual width = $2k$

Measured length = $3k \times 1.10 = 3.3k$

Measured width = $2k \times 1.12 = 2.24k$

Step 4: Compute actual and measured perimeters

Actual perimeter = $2(3k+2k) = 10k$

Measured perimeter = $2(3.3k+2.24k) = 11.08k$

Step 5: Compute the percentage error in perimeter

$$\text{Error} = (11.08k - 10k) / 10k \times 100 = 10.8\%$$

Answer: $e_1 = 10$, perimeter error = 10.8%

25

FYQ C

x is a multiple of 10. When x is increased by $x\%$ of itself, the result is a perfect square. Find the unique value of x (among multiples of 10 up to 1000).

STEPWISE SOLUTION

Step 1: Set up the equation

$$x(1 + x/100) = \text{perfect square}$$

Step 2: Test $x = 80$

$$80 \times 1.80 = 144$$

Step 3: Check if 144 is a perfect square

$$144 = 12^2 \checkmark$$

Step 4: Confirm uniqueness

A systematic check of all multiples of 10 from 10 to 1000 confirms no other value satisfies this condition.

Answer: $x = 80$

26

FYQ E

A quantity undergoes two successive increases of $p\%$ and $(p+10)\%$. Had the order of the two increases been reversed, the final value would have remained unchanged. However, the average of the intermediate values obtained in the two possible orders exceeds the original quantity by exactly 25%, while the common final value exceeds the original

quantity by exactly 56%. Find p .

STEPWISE SOLUTION

Step 1: Set up the original quantity

Let the original quantity = 100.

Step 2: Find the intermediate value for Order 1 ($p\%$ first, then $(p+10)\%$)

$$\text{Intermediate}_1 = 100 \times (1 + p/100) = 100 + p$$

Step 3: Find the intermediate value for Order 2 ($(p+10)\%$ first, then $p\%$)

$$\text{Intermediate}_2 = 100 \times (1 + (p+10)/100) = 110 + p$$

Step 4: Use the average-of-intermediates condition

$$\text{Average} = [(100+p) + (110+p)] / 2 = 105 + p$$

This average exceeds 100 by 25%, so:

$$105 + p = 125$$

$$p = 20$$

Step 5: Verify using the final-value condition

$$\text{Final value} = 100 \times (1+p/100) \times (1+(p+10)/100)$$

$$= 100 \times 1.20 \times 1.30$$

$$= 156$$

This is a 56% increase over 100, matching the given condition ✓

Answer: $p = 20$

27

FYQ C

A city's population consists of males and females, with the total population initially equal to 12,000. During the following year, the male population increases by 10% and then receives an additional net immigration of 600 people, while the female population increases by 10% and then loses 200 people through net emigration. After these changes, the ratio of males to females becomes exactly 9:8. Find the original number of males.

STEPWISE SOLUTION

Step 1: Set up variables

Let original males = M , so original females = $12,000 - M$

Step 2: Express the final male and female populations

$$\text{Final males} = 1.1M + 600$$

$$\text{Final females} = 1.1(12,000 - M) - 200$$

Step 3: Set up the 9:8 ratio equation

$$(1.1M + 600) / [1.1(12,000 - M) - 200] = 9/8$$

Step 4: Cross-multiply and simplify

$$8(1.1M + 600) = 9[1.1(12,000 - M) - 200]$$

$$8.8M + 4,800 = 117,000 - 9.9M$$

$$18.7M = 112,200$$

Step 5: Solve for M

$$M = 6,000$$

Step 6: Verify

Males = $6,600 + 600 = 7,200$; Females = $6,600 - 200 = 6,400$

Ratio = $7,200 : 6,400 = 9 : 8$ ✓

Answer: 6,000

28

FYQ B

A quantity is subjected to three successive reductions. The first reduction is 20%, the second reduction is $d\%$, and the third reduction is $(d + 10)\%$. If the overall reduction is exactly 55.2%, find d .

STEPWISE SOLUTION**Step 1: Set up the original quantity and first reduction**

Let original quantity = 100. After the first 20% reduction: $100 \times 0.8 = 80$

Step 2: Set up the retained-fraction equation for the remaining two reductions

$80 \times (1 - d/100) \times (1 - (d+10)/100) = 44.8$ (since 55.2% overall reduction leaves 44.8%)

Step 3: Simplify using $x = d/100$

$$(1 - x)(0.9 - x) = 0.56$$

$$x^2 - 1.9x + 0.34 = 0$$

Step 4: Factor and solve

$$(x - 0.2)(x - 1.7) = 0$$

$$x = 0.2 \text{ or } x = 1.7$$

Step 5: Select the valid solution

A reduction of 170% is not meaningful, so $d = 20\%$

Answer: $d = 20\%$

29

FYQ D

A committee has 500 seats, of which 40% are reserved for Category R. Within Category R, 25% of the seats are further reserved for Category S. Category S fills 60% of its allotted seats. Of the remaining unfilled S seats, 75% are transferred to the general pool, while the rest remain reserved. Separately, 40% of the Category R seats outside the S quota remain unfilled, and half of these unfilled seats are also transferred to the general pool. Find the final number of seats in the general pool.

STEPWISE SOLUTION**Step 1: Find Category R's seats and Category S's sub-quota**

Category R = 40% of 500 = 200 seats

Category S = 25% of 200 = 50 seats

Step 2: Find filled and unfilled S seats

S filled = 60% of 50 = 30; unfilled = $50 - 30 = 20$

Step 3: Transfer unfilled S seats to the general pool

75% of 20 = 15 seats transferred

Step 4: Find R seats outside the S quota and their unfilled portion

R outside S = $200 - 50 = 150$

Unfilled = 40% of 150 = 60 seats

Step 5: Transfer half of these unfilled R seats to the general pool

$60 \div 2 = 30$ seats transferred

Step 6: Compute the final general pool

Original general pool = $500 - 200 = 300$

Final general pool = $300 + 15 + 30 = 345$

Answer: 345

30

FYQ B

A sequence is defined by $a_1=100$, $a_2=110$, and for $n \geq 3$: $a(n) = a(n-1) \times (1 + a(n-2)/100)$. Find a_4 .

STEPWISE SOLUTION

Step 1: Compute a_3

$a_3 = a_2 \times (1 + a_1/100) = 110 \times (1 + 100/100) = 110 \times 2 = 220$

Step 2: Compute a_4

$a_4 = a_3 \times (1 + a_2/100) = 220 \times (1 + 110/100) = 220 \times 2.10$

Step 3: Calculate

$a_4 = 462$

Answer: $a_4 = 462$

31

FYQ A

A tank has a capacity of 20,000 litres and initially contains water equal to 40% of its capacity. At the beginning of every hour, a pump adds water equal to 20% of the tank's capacity. Immediately afterwards, one-fourth of the water present in the tank is drained. Finally, before the next hour begins, an automatic control system adjusts the water level by adding or removing an amount equal to 20% of the difference between the current water level and half the tank's capacity, whichever is required to move the water level towards half-capacity. After how many complete hours will the amount of water in the tank first exceed 10,500 litres?

STEPWISE SOLUTION

Step 1: Set up initial values

Capacity = 20,000 L

Initial water = $40\% \times 20,000 = 8,000$ L

Pump addition each hour = $20\% \times 20,000 = 4,000$ L

Half-capacity target = 10,000 L

Step 2: Compute Hour 1

Pump adds: $8,000 + 4,000 = 12,000$

Drain 1/4: $12,000 - 3,000 = 9,000$

Control adjustment: difference = $9,000 - 10,000 = -1,000$ (below half); adjust by 20% of 1,000 = 200 (add)

Water after Hour 1 = $9,000 + 200 = 9,200$

Step 3: Compute Hour 2

Pump adds: $9,200 + 4,000 = 13,200$

Drain 1/4: $13,200 - 3,300 = 9,900$

Control adjustment: difference = $9,900 - 10,000 = -100$; adjust by 20 (add)

Water after Hour 2 = $9,900 + 20 = 9,920$

Step 4: Compute Hour 3

Pump adds: $9,920 + 4,000 = 13,920$

Drain 1/4: $13,920 - 3,480 = 10,440$

Control adjustment: difference = $10,440 - 10,000 = +440$; adjust by 88 (remove)

Water after Hour 3 = $10,440 - 88 = 10,352$ (still below 10,500)

Step 5: Compute Hour 4

Pump adds: $10,352 + 4,000 = 14,352$

Drain 1/4: $14,352 - 3,588 = 10,764$

Control adjustment: difference = $10,764 - 10,000 = +764$; adjust by 152.8 (remove)

Water after Hour 4 = $10,764 - 152.8 = 10,611.2$ (exceeds 10,500)

Answer: After 4 complete hours

32

FYQ D

A trader evaluates the movement of a currency over two months. The first month's percentage gain exceeds the second month's percentage loss by 8 percentage points. Believing that gains and losses combine arithmetically, the trader expects the exchange rate to rise by 8% overall. The realised increase, however, is only 5.6%. Assuming the exchange rate changes only once in each month, find the first month's percentage appreciation.

STEPWISE SOLUTION

Step 1: Define the two rates

Let first month's gain = $a\%$.

Since the gain exceeds the loss by 8 points, second month's loss = $(a-8)\%$.

Step 2: Set up the actual (multiplicative) net change equation

$$(1 + a/100) \times (1 - (a-8)/100) = 1.056$$

Step 3: Simplify the equation

$$(100+a)(108-a) / 10000 = 1.056$$

$$(100+a)(108-a) = 10,560$$

Step 4: Expand and form a quadratic

$$10,800 + 8a - a^2 = 10,560$$

$$-a^2 + 8a + 240 = 0$$

$$a^2 - 8a - 240 = 0$$

Step 5: Solve the quadratic

$$\text{Discriminant} = 64 + 960 = 1,024 = 32^2$$

$$a = (8 + 32)/2 = 20 \text{ (rejecting the negative root)}$$

Step 6: Verify

First month: +20%; second month: $(20-8)\% = 12\%$ loss

Net multiplier = $1.20 \times 0.88 = 1.056 \rightarrow +5.6\%$ ✓

Answer: First month's appreciation = 20%

33**FYQ C**

In a survey of 100 students, 60% study Physics, 50% study Chemistry, and 40% study Mathematics. Exactly 10% of all students study all three subjects, and exactly 15% study none of the three. Find the percentage of students who study EXACTLY TWO of the three subjects.

STEPWISE SOLUTION**Step 1: Find the "at least one" percentage**

At least one = $100 - 15$ (none) = 85%

Step 2: Set up variables for exactly-one, exactly-two, exactly-three

Let a = exactly-one, b = exactly-two, c = exactly-three (given $c = 10$)

$$a + b + c = 85$$

Step 3: Use the individual-subject-sum identity

Sum of individual percentages counts exactly-one once, exactly-two twice, exactly-three thrice:

$$a + 2b + 3c = 60 + 50 + 40 = 150$$

Step 4: Subtract the two equations

$$(a + 2b + 3c) - (a + b + c) = 150 - 85$$

$$b + 2c = 65$$

Step 5: Solve for b

$$b + 2(10) = 65$$

$$b + 20 = 65$$

$$b = 45$$

Answer: 45%

34**FYQ E**

Three employees together receive the entire salary budget. The highest-paid employee earns exactly as much as the other two combined. The middle employee's salary exceeds the lowest-paid employee's salary by ₹3 lakh, and the total salary budget is ₹18 lakh. The three employees receive raises of 10%, 15%, and $r\%$, respectively. If the overall salary budget increases by 20%, determine r .

STEPWISE SOLUTION**Step 1: Set up the salary equations**

Let Lowest = L , Middle = M , Highest = H .

$$H = L + M \text{ (highest equals the other two combined)}$$

$$M = L + 3$$

$$H + M + L = 18$$

Step 2: Solve for L, M, H

$$\text{Since } H = L + M, \text{ total} = 2(L + M) = 18 \rightarrow L + M = 9 \rightarrow H = 9$$

$$M = L + 3, \text{ and } L + M = 9 \rightarrow L + (L + 3) = 9 \rightarrow 2L = 6 \rightarrow L = 3$$

$$M = 3 + 3 = 6$$

So: Lowest = ₹3 lakh, Middle = ₹6 lakh, Highest = ₹9 lakh

Step 3: Assign raises in the order the employees were introduced

The problem introduces the highest-paid employee first, then the middle, then (implicitly) the lowest — so the raises 10%, 15%, $r\%$ apply to Highest, Middle, Lowest respectively.

Step 4: Set up the new total equation

$$\text{New total} = 18 \times 1.20 = 21.6$$

$$9(1.10) + 6(1.15) + 3(1 + r/100) = 21.6$$

$$9.9 + 6.9 + 3(1 + r/100) = 21.6$$

$$16.8 + 3(1 + r/100) = 21.6$$

Step 5: Solve for r

$$3(1 + r/100) = 4.8$$

$$1 + r/100 = 1.6$$

$$r/100 = 0.6$$

Answer: $r = 60\%$

35

FYQ A

Two positive numbers have a sum of 500. One is increased by 20% and the other is decreased by 20%, resulting in a difference of 160. If the percentage changes are now reversed (the first number is decreased by 20% and the second is increased by 20%), the difference between the resulting numbers is 40. Find the two numbers.

STEPWISE SOLUTION**Step 1: Set up the sum condition**

Let the two numbers be x and y , with $x + y = 500$, so $y = 500 - x$.

Step 2: Set up the first condition (x increased 20%, y decreased 20%)

$$1.2x - 0.8y = 160$$

Step 3: Substitute $y = 500 - x$

$$1.2x - 0.8(500 - x) = 160$$

$$1.2x - 400 + 0.8x = 160$$

$$2x = 560$$

$$x = 280$$

Step 4: Find y

$$y = 500 - 280 = 220$$

Step 5: Verify using the second (reversed) condition

Reversed: x decreased 20%, y increased 20%

$$0.8x - 1.2y = 0.8(280) - 1.2(220) = 224 - 264 = -40$$

|Difference| = 40 ✓ — consistent with the given information

Answer: $x = 280, y = 220$

36

FYQ D

Runner A's speed is 25% more than Runner B's speed. A gives B a head start equal to $h\%$ of the total race distance, and as a result, they finish in a dead tie. Find h .

STEPWISE SOLUTION

Step 1: Set up the tie condition

In the same time, A covers the full distance D , while B (with the head start) only needs to cover $D \times (1 - h/100)$

Step 2: Express the speed ratio in terms of distances

$$\text{speed}_A / \text{speed}_B = D / (D \times (1 - h/100)) = 1 / (1 - h/100)$$

Step 3: Substitute the given speed ratio (1.25)

$$1 / (1 - h/100) = 1.25$$

Step 4: Solve for h

$$1 - h/100 = 1/1.25 = 0.8$$

$$h/100 = 0.2$$

Answer: $h = 20\%$

37

FYQ C

Three months' rainfall deviations from normal are -10% , $+15\%$, and $x\%$, weighted by each month's NORMAL rainfall amount in the ratio $2:3:5$. If the combined weighted deviation across all three months is exactly $+5\%$, find x .

STEPWISE SOLUTION

Step 1: Set up the weighted-average equation

$$[2 \times (-10) + 3 \times (15) + 5 \times x] / (2 + 3 + 5) = 5$$

Step 2: Simplify the numerator

$$-20 + 45 + 5x = 25 + 5x$$

Step 3: Set up the full equation

$$(25 + 5x) / 10 = 5$$

$$25 + 5x = 50$$

Step 4: Solve for x

$$5x = 25$$

$$x = 5$$

Answer: $x = 5\%$

38**FYQ E**

A battery starts at an unknown charge level $L\%$. Each hour for 2 hours, it drains 15% of its *CURRENT* charge (multiplicative loss). After these 2 hours, it is charged by a *FIXED* 20 percentage-points. If the final charge level is exactly 77.8%, find L .

STEPWISE SOLUTION**Step 1: Compute the charge after 2 hours of 15% drain**

$$\text{Charge} = L \times (0.85)^2 = L \times 0.7225$$

Step 2: Add the fixed 20-point charge and set equal to the final level

$$L \times 0.7225 + 20 = 77.8$$

Step 3: Solve for L

$$L \times 0.7225 = 57.8$$

$$L = 57.8 / 0.7225$$

Step 4: Calculate

$$L = 80$$

Answer: $L = 80\%$ **39****FYQ A**

Country A exports to four destinations: B, C, D, and E. A fixed 100 units go to E. Of the *REMAINING* exports, 30% go to B, 45% go to C, and 25% go to D. Country B's total imports are 900 units, of which A's share is 20%. Find Country A's total exports T , and then find A's percentage share of Country D's imports (given D's total imports are 600 units).

STEPWISE SOLUTION**Step 1: Find A's exports to B**

$$\text{Exports to B} = 0.20 \times 900 = 180$$

Step 2: Use the 30% share to find T

Since B receives 30% of the remaining exports ($T-100$):

$$0.30(T-100) = 180$$

$$T-100 = 600$$

$$T = 700$$

Step 3: Find A's exports to D

$$\text{Exports to D} = 0.25 \times (T-100) = 0.25 \times 600 = 150$$

Step 4: Find A's percentage share of D's imports

$$\text{Share} = 150 / 600 \times 100 = 25\%$$

Answer: $T = 700$, D-share = 25%**40****FYQ D**

Three exam papers have maximum marks 200, 300, and 500 (total 1000). A student needs 45% of the TOTAL marks (450) to pass. If the student scores 60% in Paper 1 and 50% in Paper 2, find the minimum percentage needed in Paper 3 to pass overall.

STEPWISE SOLUTION

Step 1: Find the passing mark

$$\text{Passing mark} = 0.45 \times 1000 = 450$$

Step 2: Find marks scored in Paper 1

$$\text{Paper 1} = 0.60 \times 200 = 120$$

Step 3: Find marks scored in Paper 2

$$\text{Paper 2} = 0.50 \times 300 = 150$$

Step 4: Find the marks still needed from Paper 3

$$\text{Marks needed} = 450 - 120 - 150 = 180$$

Step 5: Convert to a percentage of Paper 3's maximum

$$\text{Percentage} = 180 / 500 \times 100 = 36\%$$

Answer: 36%

41

FYQ **C**

A factory processes 1000 kg of raw material. Stage 1 (cutting) wastes 12% of the material. Stage 2 (shaping) wastes 8% of what remains after Stage 1. Stage 3 (polishing) wastes a FIXED w kg. If the final usable output is exactly 780 kg, find w .

STEPWISE SOLUTION

Step 1: Apply Stage 1 (12% wasted)

$$\text{Remaining} = 1000 \times 0.88 = 880 \text{ kg}$$

Step 2: Apply Stage 2 (8% of remainder wasted)

$$\text{Remaining} = 880 \times 0.92 = 809.6 \text{ kg}$$

Step 3: Apply Stage 3 (fixed w kg wasted) and set equal to final output

$$809.6 - w = 780$$

Step 4: Solve for w

$$w = 809.6 - 780 = 29.6$$

Answer: $w = 29.6$ kg

42

FYQ **B**

A two-digit positive integer N exceeds the sum of its digits by exactly $k\%$, where k is an integer. Reversing the digits decreases the number by 9, and the tens digit is twice the units digit. Find the value of k . Further, determine how many other two-digit positive integers have the same value of k .

STEPWISE SOLUTION

Step 1: Set up the digit variables

Let $N = 10a + b$, where a = tens digit, b = units digit.

Step 2: Use the "reversal decreases by 9" condition

$$N - \text{reverse}(N) = 9$$

$$(10a+b) - (10b+a) = 9$$

$$9a - 9b = 9$$

$$a - b = 1$$

Step 3: Use the "tens digit is twice the units digit" condition

$$a = 2b$$

Step 4: Solve the two equations together

$$a - b = 1 \text{ and } a = 2b$$

$$2b - b = 1 \rightarrow b = 1, a = 2$$

$$\text{So } N = 21$$

Step 5: Verify

Reverse of 21 is 12; $21 - 12 = 9$ ✓

Tens digit (2) = $2 \times$ units digit (1) ✓

Step 6: Find k

$$\text{Digit sum of } 21 = 2+1 = 3$$

$$N \text{ exceeds digit sum by } k\%: 21 = 3 + (k/100)(3)$$

$$18 = 3k/100$$

$$k = 600$$

Step 7: Find other two-digit numbers sharing $k = 600\%$

$$\text{Condition: } N = 7 \times (\text{digit sum}), \text{ i.e., } 10a+b = 7(a+b) \rightarrow 3a = 6b \rightarrow a = 2b$$

Valid values of b (keeping $a = 2b \leq 9$): $b = 1, 2, 3, 4 \rightarrow N = 21, 42, 63, 84$

Excluding $N = 21$ itself, the others are 42, 63, 84 $\rightarrow$ 3 other numbers

Answer: $k = 600\%$, with 3 other two-digit numbers (42, 63, 84) sharing this value

43

FYQ A

Country A exports to four destinations: B, C, D, and E. A fixed 100 units go to E, and the remaining $(T-100)$ units are split 30:45:25 among B, C, and D respectively. Country B's imports are 900 units, of which A's share is 20%. Find A's total exports T , and then find A's percentage share of Country D's imports (D's total imports = 600 units).

STEPWISE SOLUTION

Step 1: Find A's exports to B

$$\text{Exports to B} = 0.20 \times 900 = 180$$

Step 2: Use the 30:45:25 split to find T

Since B receives 30% of $(T-100)$:

$$0.30(T-100) = 180$$

$$T-100 = 600$$

$$T = 700$$

Step 3: Find A's exports to D

Exports to D = $0.25 \times 600 = 150$

Step 4: Find A's percentage share of D's imports

Share = $150 / 600 \times 100 = 25\%$

Answer: T = 700, D-share = 25%

44

FYQ **D**

A company has four pay-bands with employee COUNTS in the ratio 4:3:2:1, and average pay per employee in each band in the ratio 1:2:3:4 (Band 1 has the most employees but the lowest average pay). Find Band 1's total wage bill as a percentage of the company's total wage bill.

STEPWISE SOLUTION

Step 1: Compute each band's wage bill as (employee count) $\times$ (average pay)

Band 1: $4 \times 1 = 4$

Band 2: $3 \times 2 = 6$

Band 3: $2 \times 3 = 6$

Band 4: $1 \times 4 = 4$

Step 2: Find the total wage bill

Total = $4 + 6 + 6 + 4 = 20$

Step 3: Find Band 1's percentage share

Share = $4 / 20 \times 100 = 20\%$

Answer: 20%

45

FYQ **C**

A township developer earmarks 25% of the total land for infrastructure, reserves 20% of the balance for utilities, and allocates to the clubhouse an area equal to the difference between 15% of the original tract and 5% of the land remaining after the second stage; if only 75% of the residential land is sold and the sold portion measures 91.8 hectares, find the original area of the township.

STEPWISE SOLUTION

Step 1: Set up the original area

Let original area = T hectares.

Step 2: Apply the infrastructure allocation (25%)

Remaining after infrastructure = $T \times 0.75 = 0.75T$

Step 3: Apply the utilities allocation (20% of balance)

Utilities = $0.20 \times 0.75T = 0.15T$

Remaining after utilities = $0.75T - 0.15T = 0.60T$

Step 4: Compute the clubhouse allocation

15% of original tract = $0.15T$

5% of land remaining after second stage = $0.05 \times 0.60T = 0.03T$

$$\text{Clubhouse} = 0.15T - 0.03T = 0.12T$$

Step 5: Compute residential land

$$\text{Residential land} = 0.60T - 0.12T = 0.48T$$

Step 6: Use the sold-portion condition

$$75\% \text{ of residential land sold} = 91.8 \text{ hectares}$$

$$0.75 \times 0.48T = 91.8$$

$$0.36T = 91.8$$

Step 7: Solve for T

$$T = 91.8 / 0.36 = 255$$

Answer: Original area of the township = 255 hectares

46

FYQ B

An election has 2,000 ballots, of which 5% are found invalid after the first count. Among the remaining valid votes, Candidates A, B and C receive votes in the ratio 18 : 15 : 7. Candidate C is eliminated. Of C's votes, 25% are exhausted (cannot be transferred), while the remaining votes are split between A and B in the ratio 2 : 5. By what percentage does Candidate A's final vote tally exceed Candidate B's final vote tally?

STEPWISE SOLUTION

Step 1: Find the valid votes

$$\text{Valid votes} = 2,000 \times 0.95 = 1,900$$

Step 2: Split valid votes in ratio 18:15:7

$$\text{Total parts} = 18 + 15 + 7 = 40$$

$$\text{Each part} = 1,900 / 40 = 47.5$$

$$A = 18 \times 47.5 = 855$$

$$B = 15 \times 47.5 = 712.5$$

$$C = 7 \times 47.5 = 332.5$$

Step 3: Eliminate C and remove exhausted votes

$$\text{Exhausted} = 25\% \text{ of } 332.5 = 83.125$$

$$\text{Transferable votes} = 332.5 - 83.125 = 249.375$$

Step 4: Split transferable votes in ratio 2:5 between A and B

$$\text{Total parts} = 2 + 5 = 7$$

$$\text{Each part} = 249.375 / 7 = 35.625$$

$$A \text{ gets} = 2 \times 35.625 = 71.25$$

$$B \text{ gets} = 5 \times 35.625 = 178.125$$

Step 5: Compute final tallies

$$\text{Final A} = 855 + 71.25 = 926.25$$

$$\text{Final B} = 712.5 + 178.125 = 890.625$$

Step 6: Compute the percentage by which A exceeds B

$$\text{Difference} = 926.25 - 890.625 = 35.625$$

$$\text{Percentage} = (35.625 / 890.625) \times 100 = 4\%$$

Answer: A's final tally exceeds B's by 4%

Two investments, A and B, both start at 1000. Investment A grows LINEARLY — each period, it gains a fixed 20% of its ORIGINAL value (1000). Investment B grows MULTIPLICATIVELY — each period, it gains 20% of its CURRENT value. Find the first period in which difference of investment of A and B exceed 1,00,000.

STEPWISE SOLUTION

Step 1: Find the value of Investment A

Investment A grows linearly.

It gains 20% of its ORIGINAL value of ₹1000 every period.

$$20\% \text{ of ₹1000} = ₹200$$

Therefore, after n periods:

$$A = 1000 + 200n$$

Step 2: Find the value of Investment B

Investment B grows multiplicatively.

It gains 20% of its CURRENT value every period.

Therefore, its value is multiplied by 1.2 every period.

After n periods:

$$B = 1000 \times (1.2)^n$$

Step 3: Set up the required condition

We need the difference between B and A to exceed ₹1,00,000.

So:

$$1000 \times (1.2)^n - (1000 + 200n) > 1,00,000$$

Step 4: Check the nearby periods

For n = 25:

$$B = 1000 \times (1.2)^{25}$$

$$B \approx ₹95,396.22$$

$$A = 1000 + 200(25)$$

$$A = ₹6,000$$

Difference:

$$₹95,396.22 - ₹6,000 = ₹89,396.22$$

This is less than ₹1,00,000.

Step 5: Check n = 26

$$B = 1000 \times (1.2)^{26}$$

$$B \approx ₹1,14,475.46$$

$$A = 1000 + 200(26)$$

$$A = ₹6,200$$

Difference:

$$₹1,14,475.46 - ₹6,200$$

$$= ₹1,08,275.46$$

This is greater than ₹1,00,000.

Answer: 26th period

48FYQ **D**

A survey is sent to 2000 people, with a 45% response rate. Among respondents, 60% favor Policy X. It's estimated that non-respondents favor Policy X at a rate 20 percentage points LOWER than respondents. Estimate the overall favor-X rate across ALL 2000 people (respondents plus estimated non-respondents).

STEPWISE SOLUTION**Step 1: Find the number of respondents**

$$\text{Respondents} = 0.45 \times 2000 = 900$$

Step 2: Find respondents favoring X

$$\text{Favor among respondents} = 0.60 \times 900 = 540$$

Step 3: Find the number of non-respondents

$$\text{Non-respondents} = 2000 - 900 = 1100$$

Step 4: Find non-respondents' estimated favor rate and count

$$\text{Estimated rate} = 60 - 20 = 40\%$$

$$\text{Favor among non-respondents} = 0.40 \times 1100 = 440$$

Step 5: Find the overall favor-X count and percentage

$$\text{Total favor} = 540 + 440 = 980$$

$$\text{Overall rate} = 980 / 2000 \times 100 = 49\%$$

Answer: 49%**49**FYQ **C**

A reservoir has capacity 10,000 units and starts at 80% full. Each month, it loses 25% of its CURRENT volume to evaporation, and then gains a FIXED 400 units from rainfall. Find the reservoir's level, as a percentage of capacity, after 3 months.

STEPWISE SOLUTION**Step 1: Find the starting level**

$$\text{Starting level} = 0.80 \times 10000 = 8000$$

Step 2: Compute Month 1

$$\text{Evaporation: } 8000 \times 0.75 = 6000$$

$$\text{Rainfall: } 6000 + 400 = 6400$$

Step 3: Compute Month 2

$$\text{Evaporation: } 6400 \times 0.75 = 4800$$

$$\text{Rainfall: } 4800 + 400 = 5200$$

Step 4: Compute Month 3

$$\text{Evaporation: } 5200 \times 0.75 = 3900$$

$$\text{Rainfall: } 3900 + 400 = 4300$$

Step 5: Express as a percentage of capacity

$$\text{Percentage} = 4300 / 10000 \times 100 = 43\%$$

Answer: 43%

Two positive integers X and Y satisfy $X+Y=1800$, with X a multiple of 50. X is increased by 30%, and Y is decreased by $M\%$ (a positive multiple of 5, $M \leq 80$), resulting in a combined total of exactly 1890 (5% more than 1800). This equation alone admits SIX valid (X,M) candidate pairs. Given THREE additional clues — (1) Y is greater than 700, (2) M is an even number, and (3) the absolute decrease amount on Y is a multiple of 75 — find the unique value of X .

STEPWISE SOLUTION

Step 1: Solve the core equation for all valid (X,M) pairs

$1.3X + (1800-X)(1-M/100) = 1890$, with X a multiple of 50 and M a positive multiple of 5 (≤ 80)

This yields six candidates $(X, M, \text{decrease amount})$:

$(800, 15, 150)$, $(900, 20, 180)$, $(1050, 30, 225)$, $(1200, 45, 270)$, $(1300, 60, 300)$, $(1350, 70, 315)$

Step 2: Apply Clue 1 ($Y > 700$, i.e., $X < 1100$)

Candidates satisfying this: $X = 800, 900, 1050$

Step 3: Apply Clue 2 (M is even)

Candidates satisfying this: $X = 900$ ($M=20$), 1050 ($M=30$), 1300 ($M=60$), 1350 ($M=70$)

Step 4: Apply Clue 3 (decrease amount is a multiple of 75)

Candidates satisfying this: $X = 800$ ($150=2 \times 75$), 1050 ($225=3 \times 75$), 1300 ($300=4 \times 75$)

Step 5: Find the intersection of all three clue sets

$\{800, 900, 1050\} \cap \{900, 1050, 1300, 1350\} \cap \{800, 1050, 1300\} = \{1050\}$

Answer: $X = 1050$